



Skeleton-based human action recognition using LSTM and depthwise separable convolutional neural network

Hoangcong Le¹ · Cheng-Kai Lu¹ · Chen-Chien Hsu¹ · Shao-Kang Huang¹

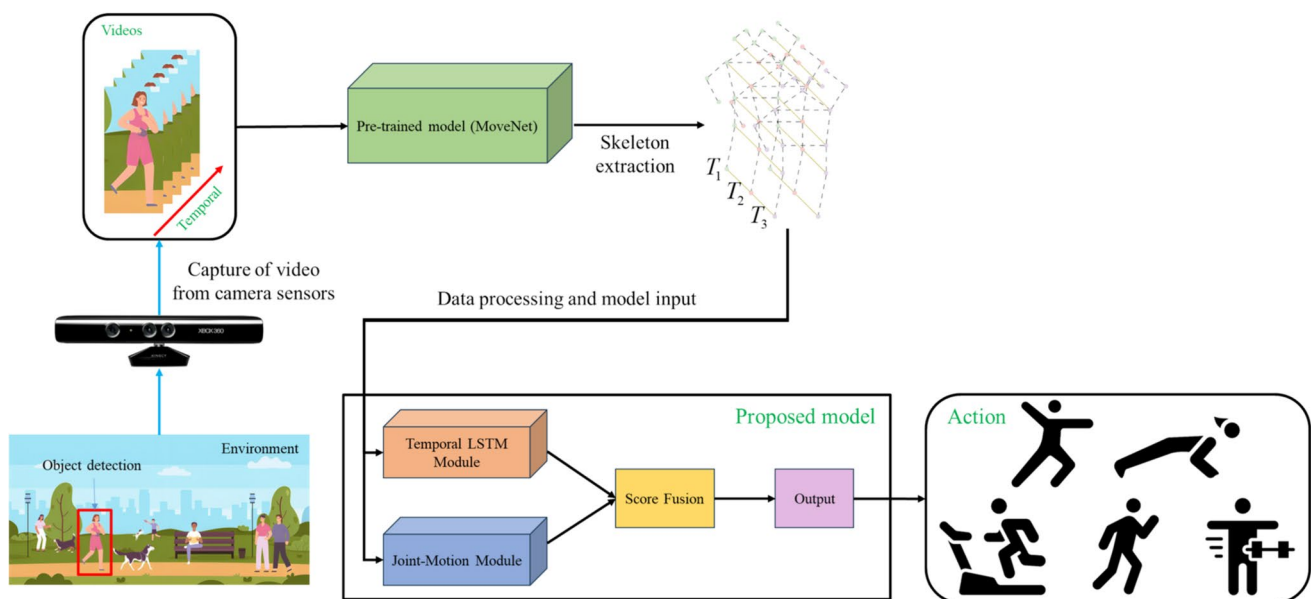
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Abstract

In the field of computer vision, the task of human action recognition (HAR) represents a challenge, due to the complexity of capturing nuanced human movements from video data. To address this issue, researchers have developed various algorithms. In this study, a novel two-stream architecture is developed that combines LSTM with a depthwise separable convolutional neural network (DSConV) and skeleton information, with the aim of enhancing the accuracy of HAR. The 3D coordinates of each joint in the skeleton are extracted using the Mediapipe library, and the 2D coordinates are obtained using MoveNet. The proposed method comprises two streams, called the temporal LSTM module and the joint-motion module, and was developed to overcome the limitations of prior two-stream RNN models, such as the vanishing gradient problem and the difficulty of effectively extracting temporal-spatial information. A performance evaluation on the benchmark datasets of JHMDB (73.31%), Florence-3D Action (97.67%), SBU Interaction (95.2%), and Penn Action (94.0%) showcases the effectiveness of the proposed model. A comparison with state-of-the-art methods demonstrates the superior performance of the approach on these datasets. This study contributes to advancing the field of HAR, with potential applications in surveillance and robotics.

Graphical abstract



Keywords Human action recognition · LSTM and DSConV architecture · 3D and 2D skeleton · Overlapping technique

1 Introduction

Human action recognition (HAR) [1] is a task in which individuals shown in videos participating in activities such as kicking a ball, walking, running, drinking water, eating rice, etc., are detected and classified. Due to its significance, HAR plays a pivotal role in various fields, including robotics, video retrieval, human-computer interaction, and surveillance. When building classification models, researchers have predominantly utilized two primary types of input data: RGB frames, and skeletal representations of humans in videos.

Prior researchers have extensively investigated the challenges associated with understanding videos and HAR [2–6], such as variations in motion and viewpoint. In recent years, scientists have utilized sequence data (RGB frames) in videos for action prediction, as each frame contains inherent spatial information about the objects and scenes shown in the image. To capture temporal information, researchers have processed sets of frames using optical flow frames to extract the features that persist over time. Ma et al. [3] proposed a method that used two streams, a spatial stream network and a temporal stream network, for feature extraction from individual images and videos, respectively. These features were concatenated, and the scientists fed them to a temporal segment LSTM or Temporal-ConvNet to predict the actions. In [4] and [5], RGB and optical flow data were used as input. The author of [4] introduced an innovative model based on 3D convolution and achieved remarkable accuracy in terms of action recognition. In [5], a method was proposed in which good frames were selected from the trimmed video domain before training and testing of a deep neural network model, thereby enhancing the performance of HAR. This approach not only reduced the computational cost but also improved the accuracy of action classification. A novel methodology [6] has also been proposed to mitigate the influence of background in videos, in which YOLOv4 is first applied to detect objects within the video frames, and the identified objects are then fed to an LSTM model to extract and learn the high-level features. This methodology has yielded superior performance in HAR systems and has been successfully applied to basketball action recognition.

The studies in [7] and [8] revealed that using a set of skeleton data for action prediction in videos, rather than relying solely on frame-based predictions, offers several advantages. This method not only reduces contextual noise and enhances robustness against background variations, but also facilitates the extraction of spatial and temporal information on the human pose. In recent research, data on human skeletons have been integrated with deep learning

algorithms, resulting in an improvement in classification accuracy. Following the significant advancements in the field of computer vision, numerous algorithms have been developed and used, such as convolutional neural networks (CNNs) [9], recurrent neural networks (RNNs) [10], long short-term memory (LSTM) [11], and graph convolutional networks (GCNs) [12], among others. Jun Liu et al. [11] demonstrated the effectiveness of combining 3D human skeleton data with an LSTM using the truth gate method, which led to outstanding performance in action classification. Hongsong Wang [10] proposed a method in which a two-stream RNN was used with 3D skeleton data to classify actions in videos. Similarly, the authors of [9] exploited the use of 3D body skeleton joints to extract additional information from videos, and then converted the 3D skeleton into tensors before feeding them to a CNN for action classification. In contrast, the researchers in [13] did not employ the 3D skeleton directly; instead, joint heatmaps were utilized, and the motion of the skeleton was leveraged to infer the activity shown in each video. Several additional models have recently been proposed for HAR in order to investigate the effectiveness of transformer-based approaches [14–17]. The RGBSformer method was introduced by the authors of [15] leveraging both RGB and skeleton information to improve the accuracy of the HAR system. Another notable model is the SpatioTemporal cRoss (STAR)-transformer proposed by Dasom Ahn et al. [16], which effectively captures multi-modal features and represents them as a recognizable vector. Furthermore, the authors of [17] utilized a transformer architecture to reconstruct 3D meshes of human bodies from RGB videos, which gave enhanced accuracy by enabling the model to learn to sample the most informative skeletal representations.

In this work, the evaluation of the proposed model's performance with both 2D and 3D skeleton data showcases its flexibility in handling different input types. Unlike previous studies [18–20] that rely on skeletons provided in original papers, two pre-trained models, Mediapipe [21] and MoveNet [22], were utilized to extract skeletons, addressing a significant gap in the analysis of different pre-trained models for skeleton extraction in HAR systems. Additionally, the use of skeletons reduces the impact of background and lighting variations in RGB frames or depth maps, while also decreasing the size of input data. The proposed model builds upon the approaches outlined in [10] and [23]. During this investigation, a vanishing gradient problem was identified in the original model and necessary modifications were implemented to improve its performance. A discussion of the challenges encountered by the original model is presented in Section 2. The proposed model consists of two streams: the first utilizes the temporal-LSTM module to establish the relationships among the joints over time,

while the second employs the joint-motion module to extract the spatial information on each skeleton and enrich temporal information. The objective was to assess the effectiveness of the pre-trained model, evaluate the quality of the videos for this task, and address potential challenges to optimizing performance for real-world applications. Furthermore, the majority of research explores the relatively uncharted combination of LSTM and depthwise separable convolutional neural networks, distinguishing it from the prevalent use of CNN, RNN, and GCN layers in improving HAR systems. The approach introduced here encompasses data processing and model creation for training and prediction of actions, as depicted in Fig. 4.

In summary, this work makes the following key contributions:

- Introduction of a novel structure combining LSTM and depthwise separable convolutional neural network (DSCoV): Previous studies [24–27] have utilized CNNs to extract temporal-spatial information from videos using either RGB frames or skeletons. In contrast, the proposed model, which incorporates LSTM and DSCoV effectively integrates skeleton data for accurate action classification.
- Feature extraction and computational efficiency: The Temporal LSTM module leverages LSTM layers to capture and model the temporal relationships between joints over time, effectively handling the dynamic nature of human motion. To complement this, the Joint-Motion module, which is based on Depthwise Separable Convolution (DSCoV), is designed to efficiently capture spatial relationships within each individual skeleton while also enriching the temporal features extracted from the sequence of skeletons. DSCoV significantly reduces the number of parameters compared to traditional convolutional models, resulting in a marked decrease in computational costs and model size. This optimization not only improves the model's execution speed but also makes it highly suitable for real-time applications [23].
- Resolution of the vanishing gradient problem [28, 29] and enhanced accuracy of the HAR system: Inspired by the work in [10] and [11], the RNN layers in the original temporal RNN were replaced with LSTM, and ideas from prior research. Additionally, the spatial RNN was transformed into a Joint-Motion module based on DSCoV, resulting in higher accuracy compared to the original model [10]. This modification enables more robust extraction of spatial-temporal information.
- SOTA performance on multiple datasets: Through the exclusive use of human pose data in the form of skeleton joints, the introduced framework achieves state-of-the-art results on the JHMDB and Florence-3D Action datasets and outperforms other approaches on the SBU Interac-

tion and Penn Action datasets. Notably, all of the experiments were conducted without the use of any tracking techniques and relied solely on human pose information.

2 Related work

In this section, a brief overview of essential knowledge underpinning this work is provided.

2.1 Information based on the skeleton

In recent years, the use of skeleton data for HAR has become increasingly popular [30–32]. Rather than relying on the rich information in each RGB frame, researchers have explored the potential of using only the human skeleton to predict actions. To obtain the joint positions and movements of the skeleton, various methods and algorithms have been proposed, including some based on deep learning [21, 33–35]. One such method is the residual steps network proposed in [33], which efficiently fuses intra-level features to carry out pose estimation for multiple individuals within each video, and extracts information about the skeletons. Yufei Xu et al. [35] developed the ViTPose model, which accurately extracts the key points of the human skeleton, resulting in high-quality pose estimation. In another method proposed in [34], called part affinity fields (PAFs), the 2D skeleton is obtained in each image and a real-time system is established for multiple-person pose detection. The study in [21] describes the use of a method of obtaining the coordinates and visibility of the skeleton. According to the authors of [21], the model obtains 33 joints, each of which has four parameters denoting its x, y, and z coordinates and visibility. The x and y coordinates are normalized to the range [0,1], whereas the z coordinate, aligned with the x-axis, represents landmark depth, using the midpoint of the hips as the origin. A smaller value of z indicates that a landmark is closer to the camera. Moreover, the visibility value, which ranges from zero to one, signifies the likelihood of the landmark being visible (i.e., present and not occluded) in the images. Researchers at Google [22] developed the MoveNet model, which detects 17 key points of a skeleton, each of which is associated with a set of (x, y, score) values. In Fig. 1, skeletons are depicted using the two methods of MoveNet and Mediapipe.

2.2 Application of deep learning to classify actions

Deep neural networks have made significant contributions to HAR. In [10], the authors developed an RNN network to recognize actions in videos, which consisted of temporal and spatial RNN components. The temporal RNN processed the 3D coordinates of the various joints in each frame,

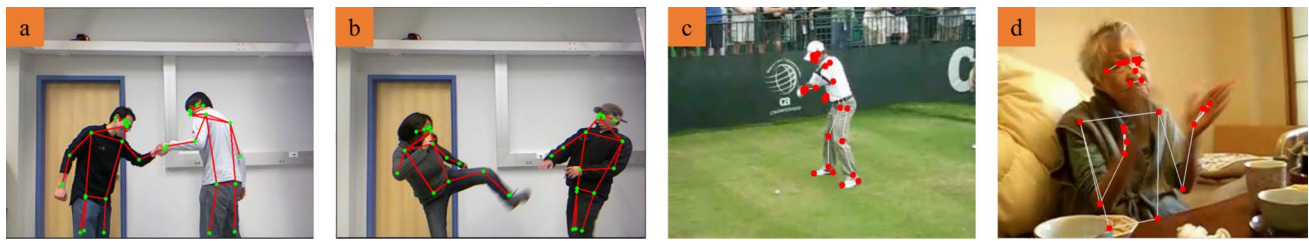


Fig. 1 Skeletons generated using MoveNet for multiple human poses contained in the SBU Interaction dataset [36]: **a** shaking hands; **b** kicking. Skeletons generated by Mediapipe for single human poses contained in the JHMDB dataset [37]: **c** playing golf; **d** clapping

producing the first score. The researchers highlighted that the human body can be perceived as an articulated system of rigid segments connected by joints. To model the spatial dependency of these joints, the graph structure was transformed into a sequence of joints and designed a corresponding RNN architecture. At each step, the RNN architecture received a vector of coordinates for a specific joint, leading to the derivation of a score for each action. These scores were combined to predict the action in the video. Similarly, in [11], an adjacency matrix was introduced by the authors to depict the relationships between the joints of a skeleton. Both the skeleton and the adjacency matrix were fed to an LSTM model and employed the truth gate model for training and action prediction. Xie et al. [12] introduced a novel method known as spatial-temporal mixing of global and local self-attention graph convolutional networks (STGL-GCN), which was based on GCN. This approach includes a local self-attention matrix to determine the strengths of connections based on the physical edges between joints.

2.3 Overview of LSTM and reason for using LSTM

RNN architectures are commonly applied to address problems involving sequence data, as RNNs are well-suited for sequence classification. LSTM is a specialized type of RNN in which three gates are used to update and control the cell states. Figure 2 illustrates the operation of an LSTM cell. The cell state, denoted as “ c ”, serves as a memory cell that

retains the information for a long period of time. The hidden state is represented by “ h ”, where h_t is the output at time step (T), and h_{t-1} and x_t are the input at time step (T). A detailed explanation of these three gates is provided in Fig. 2, elucidating their respective functionalities. In this context, LSTM aids in mapping an input composed of multiple vectors or a matrix (sequence X) to an output label Y (output data). The transition equations in an LSTM [11] are formulated as follows:

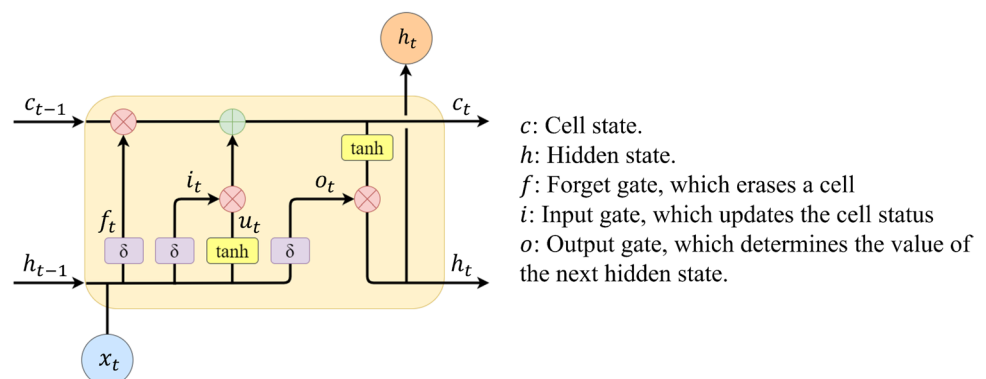
$$\begin{pmatrix} i_t \\ f_t \\ o_t \\ u_t \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} \left(M \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix} \right) \quad (1)$$

$$c_t = i_t \odot u_t + f_t \odot c_{t-1} \quad (2)$$

$$h_t = o_t \odot \tanh(c_t) \quad (3)$$

An LSTM was adopted in this work for several reasons. Firstly, as emphasized in [10], LSTM effectively addresses the issue of vanishing gradients that is commonly encountered by simple RNNs, making it particularly suitable for tasks that require long-term memory. Its incorporation of a forget gate allows selective information to be discarded from the previous cell state, effectively mitigating the vanishing gradient problem. In addition, the authors of [38]

Fig. 2 Operation of a single LSTM cell



demonstrated that LSTMs provide a temporal shortcut path, thus mitigating the issues of gradient vanishing or exploding in the temporal domain. A previous study [28] also highlighted the difficulty of training traditional RNN models to capture long-term dependencies and avoid gradient-related problems. In this research, the proposed model achieves SOTA performance on multiple datasets. A comparison of the results from the proposed model with other methodologies is presented in Section 4 to comprehensively evaluate the efficacy of the proposed approach.

3 Methodology

To address class imbalance and ensure a robust evaluation of the performance of the proposed model, a stratified k-fold cross-validation process [39] was adopted. The proposed framework, referred to here as the two-stream model, combines a temporal LSTM module with a joint-motion module. Detailed explanations are provided in this section, starting with each individual module (low level) and building up to the overall framework (high level).

3.1 Data augmentation (overlapping technique)

The overlapping technique was applied during the training process. In [40], the researchers described data augmentation as a technique for preventing overfitting by introducing variations to the training data. Overlapping is a special type of data augmentation technique that was utilized by the authors of [36] to achieve the best performance for HAR [36, 37]. In the context of HAR, the authors in [41] utilized the overlapping method to prevent overfitting and achieved an accuracy of 98.2%. Figure 3 below illustrates the operation of the overlapping method, which is adopted in Section 4 to calculate the percentage overlap for each stride, thereby augmenting the total amount of data and enabling optimal

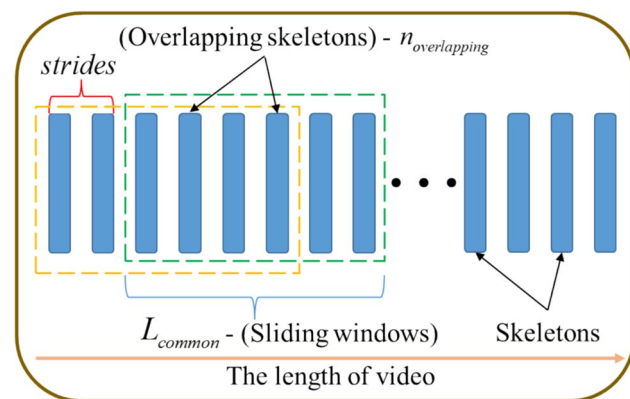


Fig. 3 Use of the overlapping technique with skeletons

results to be achieved. Here, the number of skeletons (sliding window) for the input data of proposed model is denoted as L_{common} , the number of overlapping skeletons is $n_{overlapping}$, and $strides$ refers to the number of skipped skeletons. The $percentage_overlap$ can then be calculated as indicated below:

$$percentage_overlap = \frac{L_{common} - strides}{L_{common}} \times 100\% \quad (4)$$

3.2 Architecture of the proposed model

Hongsong Wang et al. [10] demonstrated that the progression of action in a video is determined based on the sequence of skeleton images. The spatial structure refers to the coordinates and positions of the joints of each skeleton, while the temporal architecture characterizes the motion of these joints. In this paper, an end-to-end two-stream structure based on LSTM and a DSConv module is introduced. Figure 4 illustrates the data processing and the proposed approach. Within this framework, two Softmax class posteriors from two neural networks are amalgamated, denoted here as $first_score$ and $second_score$. Ultimately, the model predicts actions in videos based on the $final_score$.

Temporal LSTM module To train the first stream (temporal LSTM), each human skeleton was extracted using MediaPipe from two benchmark datasets (Penn Action [42], and Florence-3D Action [43]), as mentioned previously. MoveNet was used to obtain the multiple human poses in each video from the JHMDB and SBU interaction datasets [44, 45]. Once the coordinates of the joints in each skeleton had been obtained, $S^1 = \{P_1^1, P_2^1, \dots, P_K^1\}$ was defined as the skeleton joints for a human at time step 1 (T_1), where the joints are represented by $P_1^1, P_2^1, \dots, P_K^1$ and K is the number of joints in each human pose. The first input in Fig. 4 was then constructed as the sequence of the skeleton with n time steps, where $T = \{T_1, T_2, \dots, T_n\}$ and $T_1, T_2, \dots, T_n$ represent time steps 1, 2, ..., n , respectively. In essence, each row of the initial input corresponds to the skeleton at each time step. This input was fed to the temporal LSTM stream for training and testing. For a better comprehension of the processing using LSTM, reference Fig. 5a, which gives information on three LSTM cells for a skeleton (x_2) at input time step 2 (T_2). The LSTM model enables the creation of relationships between the joints over time. For example, P_1^1 has a relationship with P_2^1 , and P_2^1 has a relationship with P_3^1 as illustrated in Fig. 5b. The first stream is denoted as the temporal LSTM stream. When the first stream has completed processing, the value of $first_score$ is obtained. To further enhance the accuracy of the model, a second stream is then applied, as described below.

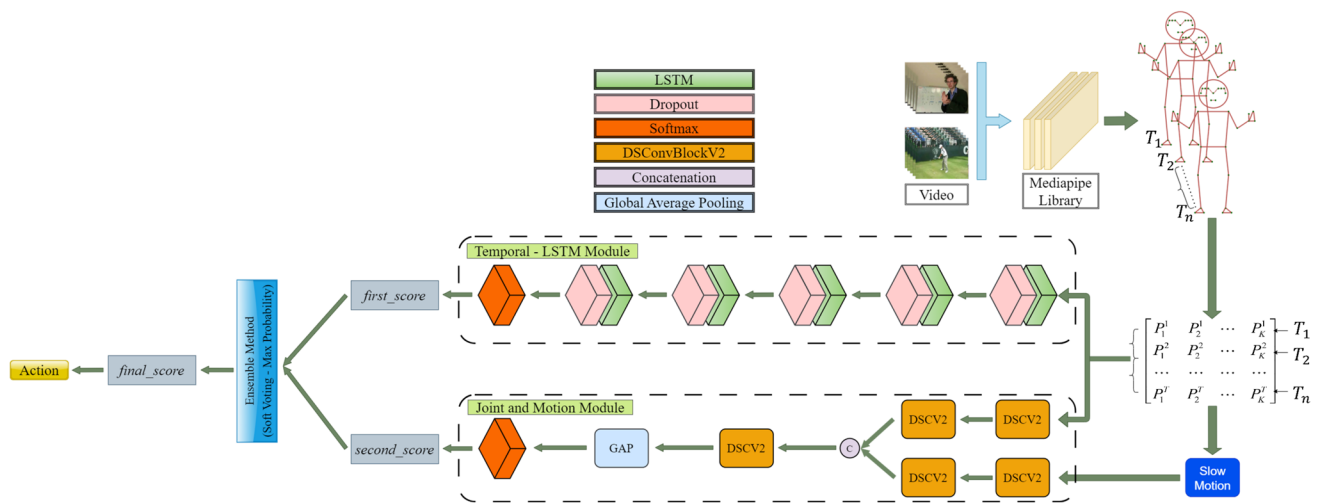


Fig. 4 Process used by the proposed model, which consists of two streams, a temporal LSTM and a joint-motion module. Here, *Softmax* refers to a fully connected layer with a Softmax activation function

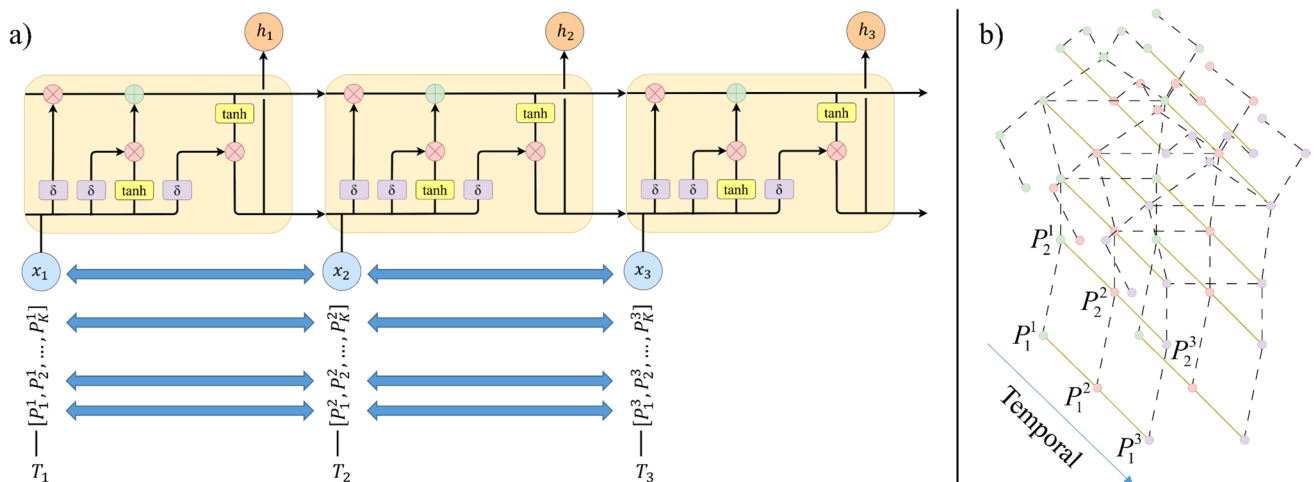


Fig. 5 a LSTM model with three LSTM cells; b relationships between joints over time, calculated using the LSTM module

Joint-motion module The authors of [23] demonstrated that traditional deep CNNs usually incur massive computational costs due to the high number of convolutions. To mitigate this, the researchers in [46] and [47] introduced DSConV, a convolutional operation that reduces the number of model parameters, as it applies different convolution kernels to extract features for each channel. However, this approach focuses solely on extracting a single aspect of the features for a specific channel. To address this limitation, pointwise convolution, a convolution operation, is employed to generate new features by calculating linear combinations of the input channels. Note that skeletons in each frame are flattened into a one-dimensional vector and fed to the block.

In addition, a comparison in [23] highlights that DSConV offers significant advantages over standard convolution (CNN) in terms of parameter reduction and computational efficiency. DSConV splits the convolution operation into two parts—depthwise and pointwise convolutions—which together reduce the computational cost by approximately 50%. The cost for DSConV can be expressed as:

$$\text{Cost}_{\text{DSConV}} = (L_k \times L_k \times C_{in} \times H \times W) + (C_{in} \times C_{out} \times H \times W) \quad (5)$$

Compared to the cost for standard convolution:

$$Cost_{CNN} = (L_k \times L_k \times C_{in} \times C_{out} \times H \times W) \quad (6)$$

This reduction is particularly beneficial for real-time applications, as DSConV significantly lowers the number of operations while maintaining effective feature extraction.

Whereas the temporal LSTM stream focuses on capturing temporal data, the second module focuses on spatial data and enhances the temporal information, which requires a dedicated approach. Introducing the DSConVBlockV2 block (Fig. 6), derived from DSConVBlock

in [23], aims to achieve optimal performance, as demonstrated in Section 4. In essence, P_1^1 has a relationship with P_2^1 and P_{K-1}^1 has a relationship with P_K^1 , as illustrated in Fig. 7b.

Slow-motion information is extensively utilized in the scheme to extract more informative global motion features and to enrich the temporal information, as illustrated in Fig. 7a. Note that L_{common} is the number of skeletons (sliding window) for the input data of the proposed model. The calculation of the slow motion (M_{slow}^T) from [23] is as follows:

Fig. 6 **a** Configuration of DSConVBlock [23]; **b** configuration of the proposed block (DSConVBlockV2)

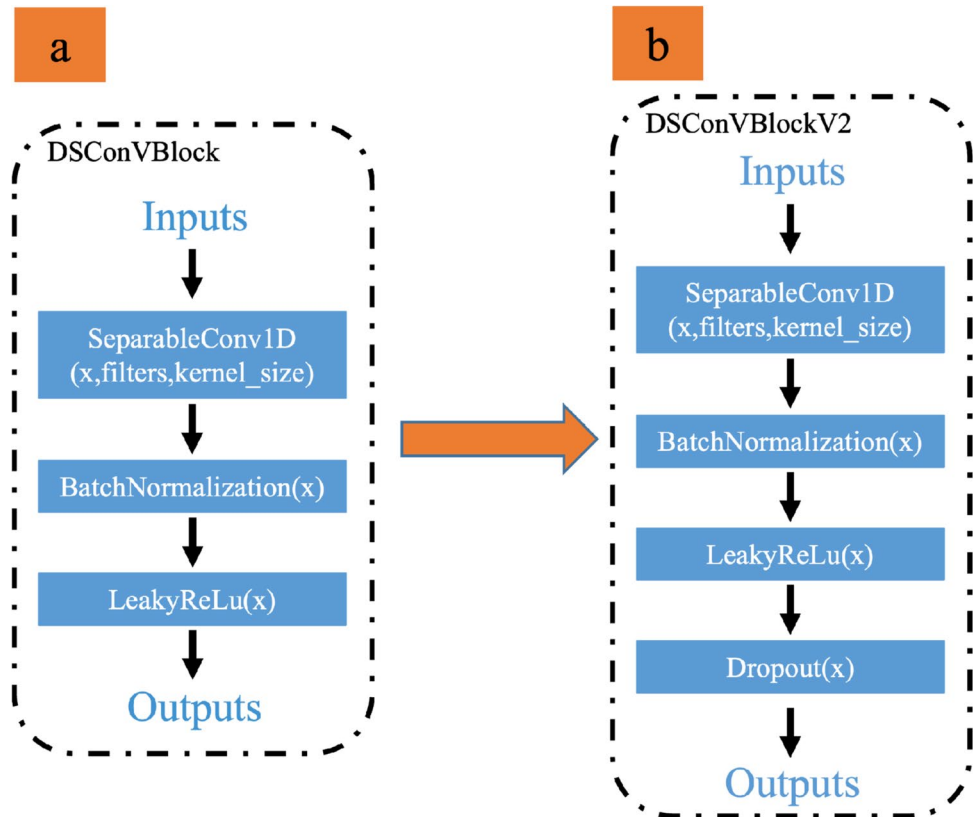
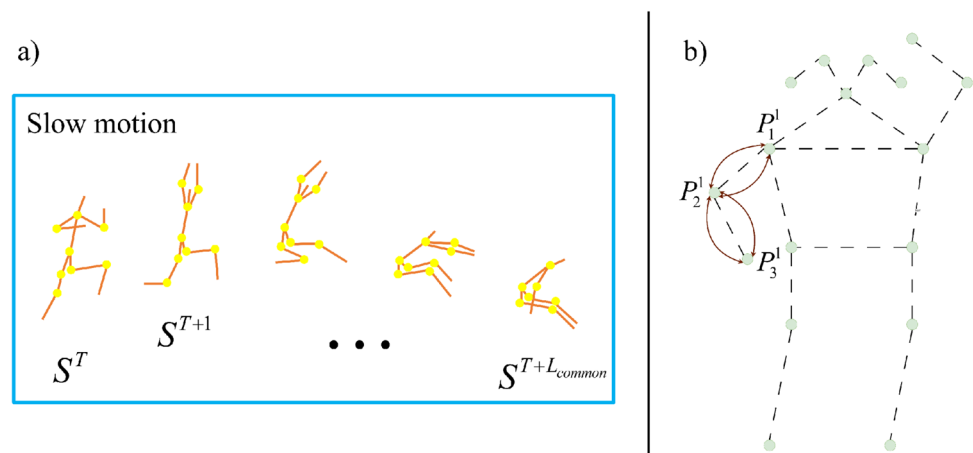


Fig. 7 **a** Simulation of slow-motion movement; **b** relationships between joints in the skeleton



$$M_{slow}^T = S^{T+1} - S^T, T \in \{1, 2, 3, \dots, L_{common} - 1\} \quad (7)$$

where M_{slow}^T denotes the slow motion at time step T , and S^{T+1} represents the next skeleton behind the S^T . After obtaining the slow-motion input, the slow-motion input is combined with the skeleton input and fed to the second (Joint-Motion) module, as illustrated in Fig. 4. This integration enables the model to generate *second_score*.

Overall structure (two-stream model) The proposed approach integrates two streams based on LSTM and DSConV networks. To address the vanishing gradient issue, a dropout layer is applied after each LSTM layer and within DSConVBlockV2. The skeleton input (S^T) is obtained using specific libraries, and the slow-motion input (M_{slow}^T) is computed using Eq. (7). It is noted that the two streams operate independently. Each stream is trained separately, and these are then combined to construct the final model.

First, the temporal LSTM module is trained to evaluate the performance of the first stream, denoted as $F_1()$. The initial input provided to this model is processed, and yields *first_score*, which is computed using the formula below:

$$first_score = F_1(S^T) \quad (8)$$

Following this, training and testing of the joint-motion module are conducted, denoted as $F_2()$. Both the skeleton and slow-motion inputs are used in this stream. The resulting output is *second_score*, which is calculated as shown below:

$$second_score = F_2(S^T, M_{slow}^T) \quad (9)$$

Next, an ensemble method is adopted, based on soft voting using the max probability approach. A combination of *first_score* and *second_score* is obtained as shown in Eq. (10) and is used to determine the value of the final score. This process involves computing the maximum probability score for each class across classifiers for every instance in the test set [48]. The *final_score* is then identified as the maximum score among these computed scores. Mathematically, it can be expressed as follows:

$$final_score = \text{Max}(first_score, second_score) \quad (10)$$

Loss function The authors of [49] and [50] highlighted the categorical cross-entropy as an effective loss function for addressing single-label categorization problems. This approach is particularly advantageous in classification tasks where only one outcome can be correct. The mathematical equation for categorical cross-entropy is given below [50].

$$\mathcal{L}_{CCE} = - \sum_{i=1}^n t_i \cdot \log_2(p_i) \quad (11)$$

The authors of [51] also demonstrated that where p_i is a probability distribution from the model's prediction, t_i is the ground truth value and n is the number of classes in each dataset.

To train the proposed model, a total loss function, $\mathcal{L}_{Total}$, is used, which consists of three loss functions: $\mathcal{L}_{Skeleton}$, $\mathcal{L}_{Temporal}$, and $\mathcal{L}_{Joint-Motion}$. Here, $\mathcal{L}_{Skeleton}$ represents the loss function for a pre-trained model such as Mediapipe or MoveNet, which enables us to obtain the skeleton, whereas $\mathcal{L}_{Temporal}$ and $\mathcal{L}_{Joint-Motion}$ correspond to the loss functions of the temporal LSTM module and the joint-motion module, respectively. For both the temporal LSTM module and the joint-motion module, categorical cross-entropy is utilized as the loss function. The overall loss function can be expressed as follows:

$$\mathcal{L}_{Total} = \mathcal{L}_{Skeleton} + \mathcal{L}_{Temporal} + \mathcal{L}_{Joint-Motion} \quad (12)$$

In this study, the aim is to minimize $\mathcal{L}_{Total}$ to obtain optimal results. $\mathcal{L}_{Skeleton}$ is fixed, as it is linked to the pre-trained model used to extract the skeleton, and the emphasis is therefore on minimizing $\mathcal{L}_{Temporal}$ and $\mathcal{L}_{Joint-Motion}$ to optimize the performance of the temporal LSTM and joint-motion modules, respectively. Minimizing both of these loss functions is therefore equivalent to minimizing $\mathcal{L}_{Total}$.

4 Experimental results

This section provides details of the benchmark datasets used in the experiments and the parameters for the experimental setup. The proposed method is compared to state-of-the-art (SOTA) methods on each dataset and demonstrates that the use of both streams outperforms the use of temporal LSTM alone.

4.1 Datasets

This section provides detailed information about each dataset, including the number of training and testing samples in each split, as illustrated in Table 1. In instances where the pre-trained model was unable to extract skeletons due to poor video quality, those videos were excluded from both the training and testing datasets.

Penn action [42] This consists of 2326 videos captured in the wild, depicting 15 different actions, including baseball pitching, bench pressing, strumming a guitar, and “clean and jerk”, among others. Many of the videos in this dataset have missing body parts, and the image scales vary significantly from one sample to another, which poses a challenge. For evaluation, three-fold cross-validation was used, and the average accuracy was computed.

Table 1 Numbers of training and testing samples in each dataset after data processing

Name of dataset	Split 1		Split 2		Split 3		Split 4		Split 5	
	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test
Penn Action [42]	6109	714	6108	712	6122	714	-	-	-	-
JHMDB [44]	2293	310	2296	308	2296	310	-	-	-	-
Florence-3D Action [43]	612	43	615	50	610	45	607	50	611	50
SBU Interaction [45]	251	57	256	57	253	56	249	56	251	56

Florence-3D action [43] This contains 250 action sequences representing nine activities performed two or three times by 10 different objects. In this dataset, the 3D coordinates of the skeleton are extracted with 33 body-joints for each person in the videos. Some actions in the dataset include answering a phone, looking at a watch, sitting down, tightening laces, and standing up. For evaluation, the dataset was divided into five folds and best accuracy was reported.

JHMDB [44] This is a subset of HMDB51 that consists of 928 short videos divided into 21 action classes. The videos in this dataset usually show only one person or a few objects, making it easier to extract the skeletons. The accuracy of this dataset was computed based on the average classification accuracy, using a standard three-fold split.

SBU interaction [45] This contains interactions between two persons captured by Kinect, making it a complex human action dataset. MoveNet was adopted to extract each skeleton, with 17 joints. The dataset includes 6822 frames with 282 skeleton sequences, where eight activities are performed by seven individuals in a laboratory environment. This dataset presents certain challenges, as the interactions are non-periodic and involve very similar body movements. For evaluation, the dataset was split into five folds and the average recognition accuracy was reported.

4.2 Implementation details

The MoveNet model was used [22] to extract the skeletons from each frame in the JHMDB and SBU Interaction datasets, as it can detect multiple (up to six) people in a video. For the remaining two datasets, the Mediapipe library was adopted [21], which can extract the skeleton of a single person in a video. The MoveNet model returns 17 joints plus the background, whereas the Mediapipe library gives 33 joints plus the background. To train the proposed model, the Adam optimizer was used, with a learning rate of $3E-3$, across all datasets, and the categorical cross-entropy loss function was applied for action classification. Moreover, a stride was applied with values of 11, nine, five, and five for the Penn Action, JHMDB, SBU Interaction, and Florence-3D Action datasets, respectively. Dropout was applied after the LSTM layers and within DSConVBlockV2, which proved

beneficial. The proposed networks were implemented using TensorFlow on a workstation equipped with two NVIDIA GeForce RTX 3090 Ti GPUs (24GB RAM each) and additional resources provided by Google Colab Pro. The model checkpoint training technique was employed to achieve the best accuracy. The two streams were trained independently, and pre-trained models were considered for evaluation on the same testing data. Table 2 shows the parameters for both streams for each dataset based in these experiments. Notably, during the investigation, the optimal sequence lengths for Penn Action, JHMDB5, SBU Interaction, and Florence-3D Action were found to be 45, 35, 20, and 20, respectively, suggesting the use of these sequence lengths to achieve the best

Table 2 Parameter settings for each module on each dataset

Penn Action [42]	$L_{common} = 45$ $strides = 11$	
	Temporal-LSTM: • Epochs = 200 • Batch_size = 128 • Dropout = 0.4 • Units = 256	Joint-motion: • Epochs = 200 • Batch_size = 128 • Dropout = 0.4 • Filters = 256 • Kernel_size = 7
JHMDB [44]	$L_{common} = 35$ $strides = 9$	
	Temporal-LSTM: • Epochs = 200 • Batch_size = 16 • Dropout = 0.4 • Units = 256	Joint-motion: • Epochs = 300 • Batch_size = 16 • Dropout = 0.4 • Filters = 256 • Kernel_size = 7
Florence-3D Action [43]	$L_{common} = 20$ $strides = 5$	
	Temporal-LSTM: • Epochs = 200 • Batch_size = 16 • Dropout = 0.3 • Units = 128	Joint-motion: • Epochs = 300 • Batch_size = 16 • Dropout = 0.3 • Units = 256 • Kernel_size = 7
SBU Interaction [45]	$L_{common} = 20$ $strides = 5$	
	Temporal-LSTM: • Epochs = 200 • Batch_size = 16 • Dropout = 0.3 • Units = 128	Joint-motion: • Epochs = 300 • Batch_size = 16 • Dropout = 0.3 • Units = 256 • Kernel_size = 7

Table 3 Results for each module and the proposed method on the JHMDB dataset

Model	Split 1	Split 2	Split 3	Mean accuracy
Temporal-LSTM	68.06	61.15	64.26	64.49
Joint-Motion	75.67	67.99	75.67	72.93
Two-stream	75.67	68.71	76.05	73.48

Table 4 Comparison of accuracy with state-of-the-art methods on JHMDB

Method	Year	Features used	Accuracy
EHPI [13, 56]	2019	P	60.5
DynaMotion [13, 57]	2020	P	60.2 ⁺
JMRN [13]	2022	P*	68.5
Bi-LSTM [52]	2022	P*	70.2
POSEII [53]	2023	P	70.1
HAR-depth [54]	2021	Ds + RGBs	73.1
TP [55]	2024	CHs + OFs + RGBs	69.9
Proposed model	2024	P*	73.5

Models marked * use difference features along with standard heatmaps, and include an ensemble pose model [13]. + indicates results from the first split alone [13]

P human pose (pose heatmaps) [13], PAF part affinity fields [13], CF features from the pose extractor CNN [13], P* human pose (skeleton data – 2D or 3D), Ds depth images, CHs colorized heatmaps, OFs optical flow

results with the scheme. The numbers of DSConVBlockV2 and LSTM layers for each stream are depicted in Fig. 4. Data augmentation techniques were applied to normalize the data and ensure uniform input lengths across sequences. During the testing phase, final frames from the original input data were duplicated and appended to subsequent frames to maintain consistent sequence lengths.

4.3 Results on JHMDB dataset

Table 3 shows the accuracy for each stream and the overall accuracy of the two-stream model based on LSTM and DSConV for each individual split, to enable a comparative analysis between the individual streams and the proposed model. The two-stream model outperformed the joint-motion module on two splits (splits 2 and 3) with advantages of +0.72% and +0.38%, respectively. Furthermore, the proposed approach achieved higher accuracy than both the temporal LSTM module and the joint-motion module on all splits. The average accuracy for the method stands at 73.48%, surpassing that of the temporal LSTM (64.49%) and the joint-motion (72.93%) models.

In Table 4, the two-stream model is compared to several baseline methods using human poses as input, using

human poses as input, demonstrating favorable good results (73.5%). Compared to JMRN [13], Bi-LSTM [52], and POSEII [53], the proposed approach gives improvements in accuracy of about 5%, 3.3%, and 3.4%, respectively. In [52], the authors utilized the PCA algorithm for dimensionality reduction of input data and augmented their method with Bi-LSTM to enhance information retention from both past and future directions, yet their model's accuracy remained lower than that of the proposed approach. This disparity may stem from information loss due to the PCA algorithm. In 2023, the authors of [53] carried out human pose estimation with ResNet and a temporal shift module (TSM) on the JHMDB dataset. ResNet was used to extract features from the input data, while TSM learned temporal information. However, the accuracy of this method (70.1%) was still lower than that of the proposed model (73.5%), as the proposed approach exploits several kinds of input data, such as the original human skeleton and slow motion, whereas POSEII only considers the original human pose. Furthermore, the proposed model is designed with LSTM layers, which enable it to learn temporal information more deeply compared to TSM, which is based on 1D-CNN. The method proposed here also outperformed several state-of-the-art methods that combines multiple types of input data, such as HAR-Depth (73.1%) [54] and TP (69.9%) [55]. Although TP [55] also utilizes various types of input data, such as CHs, OFs, and RGBs, its accuracy was still lower than the proposed method, as TP [55] is implemented using CNN layers, which have a reduced ability to learn from long sequences over time compared to LSTM layers. DD-Net [51] achieved a higher accuracy than the proposed method, a finding that can be attributed to the use of additional input data such as joint collection distances, slow motion, and fast motion in addition to the original 2D skeleton data.

These results indicate that the proposed two-stream model effectively exploits the temporal-spatial information of the skeleton to achieve a high accuracy for HAR. Section 2 highlights the advantages of RNN and LSTM layers when applied to sequence data. However, the authors of [13] used CNN layers to generate a neural network rather than RNN or LSTM layers, and this key difference is one of the reasons why the proposed framework performs better than JMRN [13].

4.4 Results on Florence-3D Action dataset

This section, discusses the outstanding performance of the method and compares it with alternative methods on Florence-3D Action, in line with the findings presented in [9]. Table 5 shows a comparison of the best results, following the experimental design outlined in [15]. Additionally, some drawbacks of the proposed framework for HAR are

Table 5 Comparison of the proposed approach against state-of-the-art models

Model	Year	Accuracy
SCK [9]	2022	92.98
DCK [9]	2022	92.77
SCK + DCK [9]	2022	95.23
SCK $\oplus$ [9]	2022	96.50
DCK $\oplus$ [9]	2022	96.41
SCK $\oplus$ + DCK $\oplus$ [9]	2022	97.45
Kendal Traj. [9, 58]	2022	93.04
Kernel + ResNet [9]	2022	95.40
Proposed model	2024	97.67

discussed and explain why it performs better than SCK $\oplus$ + DCK $\oplus$ [9].

Table 5 indicates that the proposed approach achieves a higher accuracy (97.67%) than the other models, such as Kernel + ResNet [9] (95.4%), and Kendall Traj [9, 58] (93.04%). In [9], two tensor-based feature representations were proposed, called SCK and DCK, to extract information and motion data on human poses by transforming them into tensors. The compatibility between two action sequences was calculated based on the similarity between the skeletons and their temporal orders, using a tensor obtained from SCK and DCK, and SVM was subsequently employed to classify the actions. As mentioned previously, LSTM in the model allows it to handle sequence data more effectively than the other algorithms, thus allowing it to outperform

SCK, DCK, and SCK + DCK [9] by about 4.69%, 4.9%, and 2.44%, respectively. The authors of [9] also improved SCK and DCK by introducing SCK $\oplus$ and DCK $\oplus$, which achieved better accuracy. SCK $\oplus$ and DCK $\oplus$ gave similar accuracies of about 96.5% and combining SCK $\oplus$ with DCK $\oplus$ resulted in a higher accuracy (97.45%), surpassing both SCK $\oplus$ and DCK $\oplus$, as the combined SCK $\oplus$ + DCK $\oplus$ model operates on subsequences to capture the local-global interplay of correlations. However, the proposed approach still outperformed the SCK $\oplus$ + DCK $\oplus$ method by 0.22%, since the joint-motion module also effectively captures the local-global information on each joint in each skeleton.

Figure 8 illustrates a limitation that arises when using skeleton data alone for HAR. As shown in Fig. 8a and b, two distinct actions, namely waving and drinking from a bottle, are shown in the Florence-3D Action dataset. However, the corresponding skeletal data, as shown in Fig. 8c and d, are very similar to each other, making it difficult to accurately predict the action solely based on skeleton data. To improve the performance of the model and address this limitation, additional inputs such as RGB or Optical Flow could be incorporated to capture more contextual information from the video background. The research conducted in [16] confirmed that combining RGB video frames with human pose data can enhance the accuracy of HAR. Another approach might involve the adoption of a subsequence-based approach, similar to that in [9], to better capture the local-global interplay of correlations. These avenues could be considered in future work to augment the accuracy of the proposed approach for HAR.

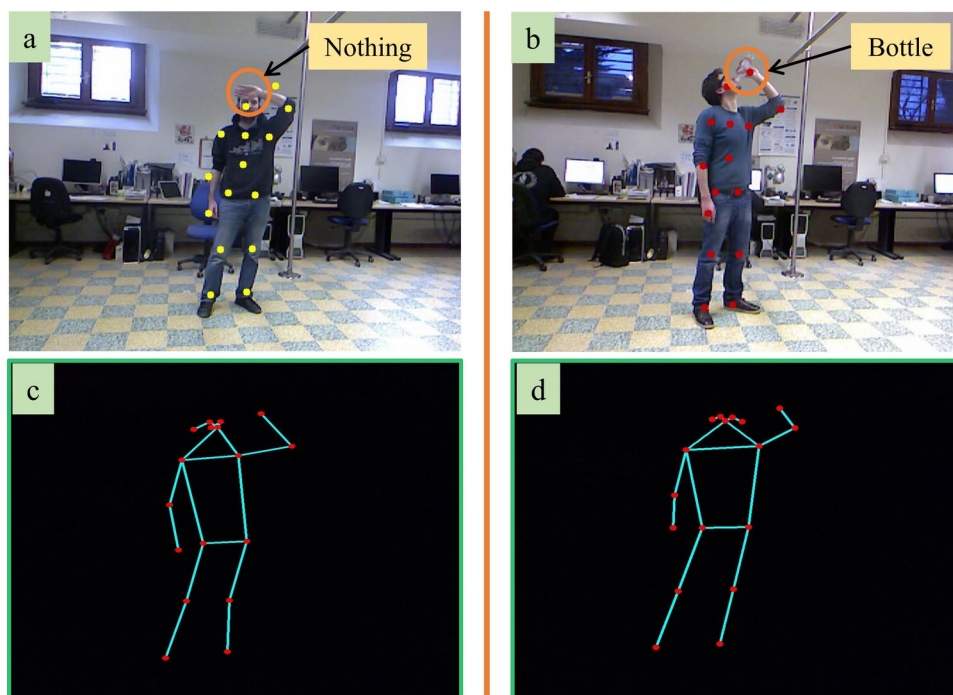
Fig. 8 Data from the Florence-3D Action dataset: **a** waving; **b** drinking from a bottle; **c** skeleton obtained for the waving action; **d** skeleton obtained for drinking from a bottle


Table 6 HAR accuracy for the proposed method and the two-stream RNN on the SBU Interaction dataset

Method	Technique	Accuracy
Temporal RNN [10]	Stacked	89.0
	Hierarchical	90.2
Spatial RNN [10]	Chaining	82.2
	Traversal	86.6
Two-stream RNN [10]	No transform	91.9
	3D transform	94.8
Temporal LSTM module	Overlapping	93.0
Joint-motion module	Overlapping	87.1
Proposed model	Overlapping	95.2

4.5 Results for SBU Interaction dataset

To further demonstrate the effectiveness of the proposed approach, it was compared with other methods, as presented in Table 6. For the first stream, which extracts temporal information, the original paper [10] employed two techniques: stacked and hierarchical schemes, resulting in accuracies of 89.0% and 90.2%, respectively. In the hierarchical approach, each skeleton was divided into five parts, and actions were performed by either an independent part or a combination of several parts. This led to the conclusion that the accuracy of a combination of the temporal RNN and a hierarchical approach was higher than for the temporal RNN with the stacked technique.

In this study, a problem arose in that it was challenging to divide each skeleton into five or several parts as in the original paper. Hence, the proposed temporal LSTM module uses a stacked and overlapping approach rather than a hierarchical technique, and an accuracy of 93% is achieved for this dataset, a value higher than for the stacked and hierarchical techniques alone by approximately 4% and 2.8%, respectively. To extract spatial information, the authors of [10] employed a spatial RNN together with techniques such as chaining and traversal to generate a graph structure in the form of a sequence of joints, yielding accuracies of about 82.2% and 86.6%, respectively. In contrast, DSConVBlockV2 was developed to achieve outstanding performance in the second stream and surpassed the chaining technique by 4.9% and the traversal technique by 0.5%. Furthermore, when both streams were combined, the proposed framework achieved an accuracy of 95.2%, outperforming the two-stream RNN approach [10] described with no transform (91.9%) or a 3D transform (94.8%).

In Table 7, the performance of the two-stream framework is compared with various baseline methods on the SBU Interaction dataset. The proposed method outperformed other the baseline methods across the different network architectures. This two-stream model based on LSTM and

Table 7 HAR accuracy for the two-stream model and existing models on the SBU Interaction dataset

Model	Year	Accuracy
Trust Gate ST-LSTM [10, 11]	2016	93.3
Two-stream RNN [10]	2017	94.8
RV + HS + GCNs [18]	2020	94.5
Discr. NBNN + PCA-15 [61]	2021	94.2
Three-stream CNN [59]	2022	94.2
Bi-LSTM [60]	2023	90.0
Proposed model	2024	95.2

DSConV outperformed a three-stream CNN [59], Bi-LSTM [60], and a two-stream RNN [10] by about 1%, 5.2% and 0.4%, respectively. The method also surpassed Bi-LSTM [60] by leveraging both the original skeleton and slow-motion inputs, enabling the extraction and learning of more high-level features. In contrast to [60], which relies solely on the original skeleton, the method benefits from incorporating slow-motion data. In addition, the method (95.2%) outperformed RV + HS + GCNs [18] (94.5%) and Discr. NBNN + PCA-15 [61] (94.2%). Although RV + HS + GCNs [18] uses GCNs to establish strong spatial relationships between the joints, the effectiveness of this approach in terms of capturing joint relationships over time is inferior to the use of LSTM layers. This weakness in regard to learning from long sequences is a primary reason for the superiority of the proposed approach over RV + HS + GCNs [18]. The incorporation of the overlapping technique within the proposed model also amplifies the amount of data extracted from the SBU Interaction dataset, meaning that more information can be gathered on the actions. This enables the proposed method to outperform the two-stream RNN. These results demonstrate the effectiveness of the two-stream model based on LSTM and DSConV on small-scale HAR datasets.

The confusion matrices depicting the performance of the temporal LSTM and joint-motion modules, and the two-stream model, are shown in Fig. 9. There are only two correctly identified actions (departing and hugging) for the joint-motion module in Fig. 9c, whereas there are three pairs of misclassified actions for temporal LSTM (kicking, exchanging objects, and punching), and only two for the two-stream model (kicking and punching). For the pushing action, the accuracy of Temporal LSTM (75%) is lower than that of the two-stream model (88%). In addition, for the action of exchanging objects, the samples are misclassified as shaking hands by 14% in both the temporal LSTM (9b) and joint-motion modules (9c), while the full model can correctly recognize all the samples. These results demonstrate that the full framework outperforms the temporal LSTM and joint-motion modules.

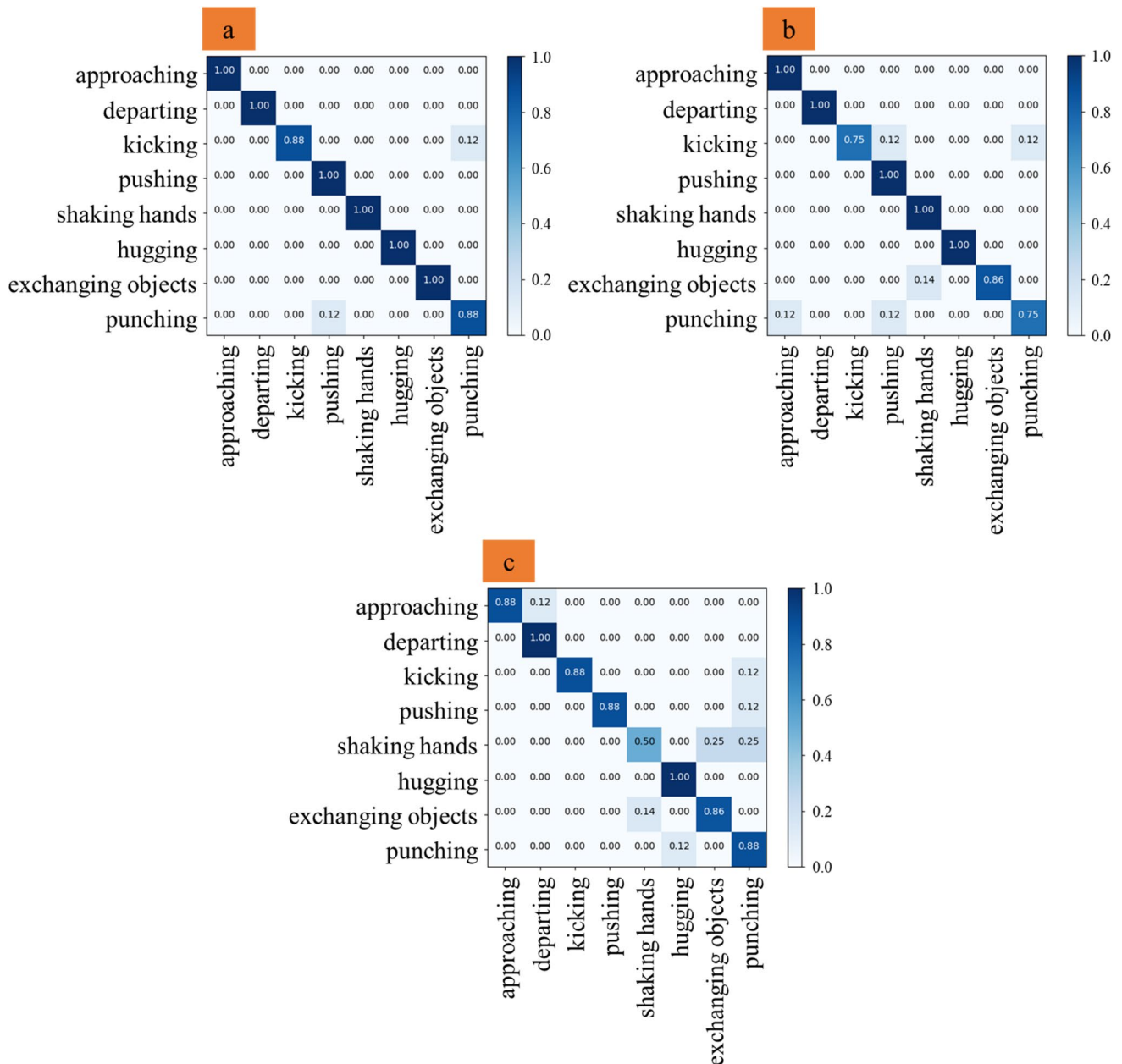


Fig. 9 Comparison of confusion matrices on the SBU Interaction dataset: **a** full two-stream model; **b** temporal LSTM module; **c** joint-motion module

4.6 Results on Penn Action dataset

Since almost all existing methods make use of a temporal RNN or temporal LSTM, the aim of this section is to illustrate the advantages of a two-stream approach over the temporal LSTM and joint-motion modules on the Penn Action dataset, as shown in Fig. 10. The red and black rectangles in the figure indicate the accuracy gap between the proposed model and the temporal LSTM or joint-motion modules, respectively, while also displaying the accuracy for each activity. The full two-stream model outperforms

the individuals' modules for certain actions in the dataset. For example, the accuracy of the two-stream model is higher than that of the temporal LSTM module by 3.3%, 4.7%, 4.2%, and 4% for the actions of bowling, pull-ups, squats, and tennis forehands, respectively. In comparison to the joint-motion module, the proposed approach achieves accuracy values that are approximately 1.6%, 1.6%, 1.4%, and 2% higher for these actions. The performance of the joint-motion module is superior to that of the temporal LSTM.

The evaluation conducted on Split 1 of the Penn Action benchmark dataset demonstrates the effectiveness

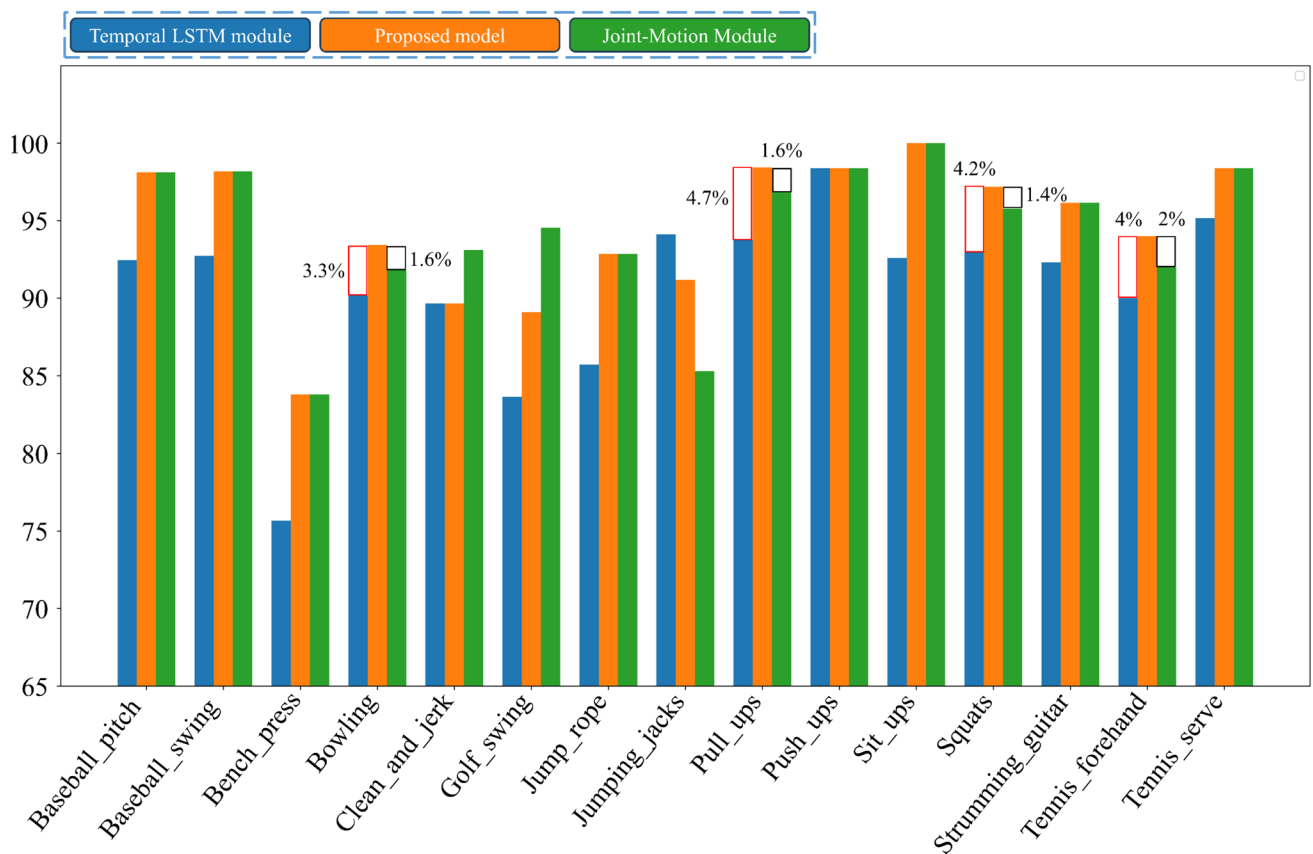


Fig. 10 Accuracy for each action in the Penn Action dataset

Table 8 Results for the proposed model on each class based on precision, recall, and F1-score

Name of class	Precision	Recall	F1-score
Baseball pitch	96.30%	98.11%	97.20%
Baseball swing	93.10%	98.18%	95.58%
Bench press	91.18%	83.78%	87.32%
Bowling	89.06%	93.44%	91.20%
Clean and jerk	96.30%	89.66%	92.86%
Golf swing	98.00%	89.09%	93.33%
Jump rope	96.30%	92.86%	94.55%
Jumping jacks	96.88%	91.18%	93.94%
Pull-ups	100.0%	98.44%	99.21%
Push-ups	95.31%	98.39%	96.83%
Sit-ups	100.0%	100.0%	100.0%
Squats	95.83%	97.18%	96.50%
Strumming guitar	89.29%	96.15%	92.59%
Tennis forehand	95.92%	94.00%	94.95%
Tennis serves	93.85%	98.39%	96.06%
Mean value	95.15%	94.59%	94.81%

of our model, as detailed in Table 8. Some actions, such as bench press, bowling, and strumming guitar, exhibited relatively lower precision values (~91.18%, 89.06%, and 89.29%, respectively), while sit-ups and pull-ups achieved the highest precision scores (100%). Notably, the performance on sit-ups was exceptional, with perfect scores across all three metrics: precision, recall, and F1-score.

There were certain challenges associated with using the Mediapipe model to extract skeletons from the videos in the Penn Action dataset. In Fig. 11a, it is evident that the pre-trained model fails to extract the skeleton in both the first and fifth frames, with inaccuracies in the shapes of the skeletons in the remaining frames deviating from the actual shape of the individual in the video. The reason is that the frames in Fig. 11a contain numerous objects, which introduce noise for the pre-trained model. In Fig. 11b, the positions of the joints in the first, second, and third frames are incorrect, probably because the videos did not capture the full body of the person.

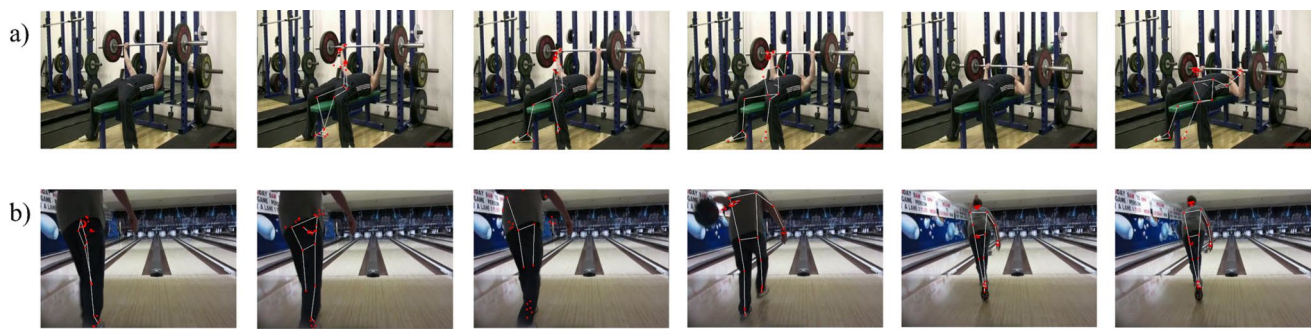


Fig. 11 Examples of mistakes when extracting skeletons from the Penn Action benchmark with Mediapipe: **a** bench pressing; **b** bowling

Table 9 Comparison of the proposed approach with SOTA methods on the Penn Action dataset

Method	Pre-training	Modality	Accuracy
2s-AGCN [63]	✓	Pose	89.5
MS-G3D [63]	✓	Pose	88.5
SKD [19]	×	Pose	81.9
Cuiwei Liu et al. [20]	×	Pose	90.5
JDD [62]	×	Pose	93.2
HDM [62]	×	Pose	93.4
Proposed approach	×	Pose	94.0

The performance of the proposed method is compared with SOTA methods in Table 9. The model (94.0%) outperforms methods based on pose features (JDD [62], MS-G3D [63], HDM [62]), and gives superior performance compared to the 2s-AGCN method [63], which is based on appearance features and used pre-training model, by about 4.5%. Both the two-stream model and the other methods were trained and tested only on the given data, with no pre-training. The SCK model developed by [19], achieved an accuracy of 81.9%, lower than that of the proposed model. This disparity arises because the scheme in [19] considers only one type of input data, specifically the original skeleton. In contrast, the introduced methodology incorporates two types of input data, enabling deeper learning of the temporal-spatial information from the original skeleton. As it leverages the advantages of LSTM layers over GCN layers, the model gives superior performance, especially when applied to long sequence data. The researchers in [20] proposed an innovative approach to enhance the performance of a HAR system using low-quality skeleton data through part-level knowledge distillation. Despite efforts to improve model learning by focusing on several parts, however, the inherently low quality of the skeleton data results in weakened temporal-spatial information in the input data. This limitation explains why

the proposed model achieved higher accuracy (94.0%) compared to the novel approach presented in [20] (90.5%). The authors of [16] demonstrated the remarkable effectiveness of the Transformer model by combining RGB and pose as input data; as a result, the STAR-Transformer achieves a higher accuracy than the proposed model.

4.7 Ablation studies

In this section, experiments are conducted on the JHMDB datasets as an example to investigate the different components of the temporal LSTM module, the joint-motion module, and the full two-stream framework. In these experiments, the sequence length for JHMDB was 35 time steps.

4.7.1 Effects of stride on temporal-LSTM

In [36], the authors split a YouTube video into multiple clips using an overlapping sliding window approach, with a window size of 100 frames and an overlap of 50 frames, representing a 50% overlap. In [37], the baseline used a sliding window with a 75% overlap on HMDB51. Hence, the experiments in the study considered overlap percentage ranging from 50 to 75%, as suggested in Eq. (4). The results are illustrated in Fig. 12. It is noted that with overlap proportions of 80% and 77% and strides of seven and eight, respectively, the specified range of 50–75% was exceeded. Consequently, for the temporal-LSTM module, experiments were conducted with stride values ranging from seven to 15. The highest accuracy of 66.16% was achieved for a stride of nine, with a percentage overlap of about 74%. Hence, an overlap percentage of around 75% was selected for the proposed approach. In HAR, the data used for training and testing can significantly influence the model's ability to learn and extract information from videos or actions. These findings underscore the effectiveness of selecting an optimal stride value.

Fig. 12 Percentage overlap and accuracy for varying values of the stride

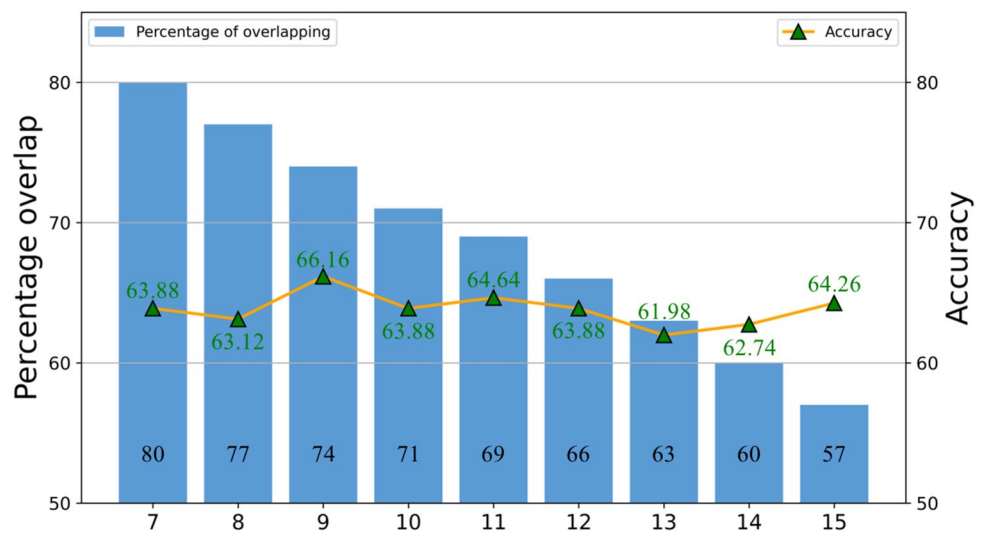
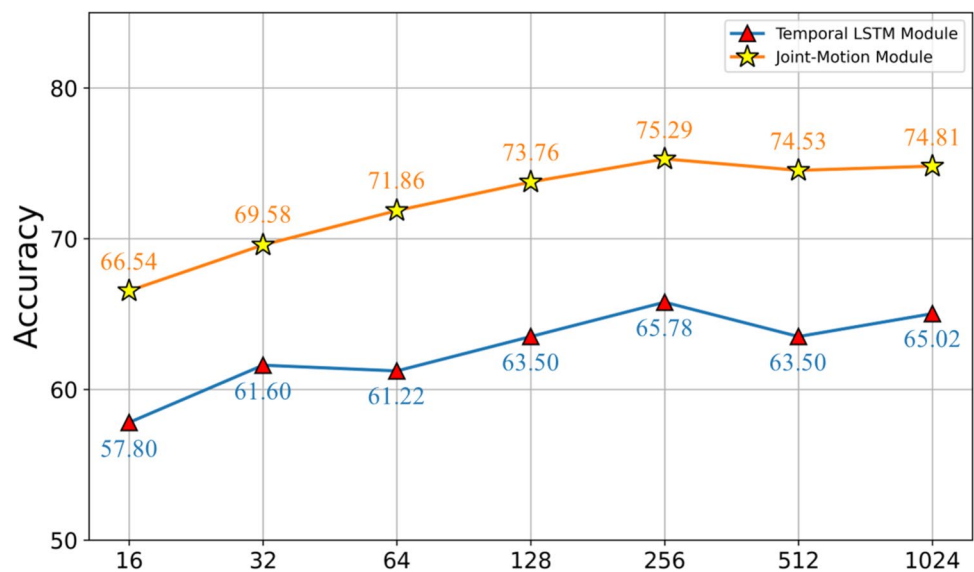


Fig. 13 HAR accuracy for the joint-motion and temporal LSTM modules with fine-tuning of the unit/filter parameters



4.7.2 Effect of the number of units/filters (positive integer, dimensionality of the output space) on each stream

To gain deeper insight into the operation of each stream, an ablation study was conducted on the number of units/filters, as shown in Fig. 13. Initially, the analysis focused on the joint-motion module. When the number of units was set to 16, 32, 64, and 128, the accuracy of the compressed model was 66.54%, 69.58%, 71.85%, and 73.76%, respectively. The accuracy decreased from 75.29 to 74.53% when the number of units was reduced from 256 to 1024. For the temporal LSTM module, the accuracy peaked at 65.78% with 256 filters. When using different numbers of units or filters, the results for the temporal LSTM and joint-motion framework were comparable, and the number of units and filters was therefore set to 256 for each stream. However,

it was observed that the optimal accuracy was achieved when the number of neurons in the stacked LSTM of the temporal LSTM was set to 128 (or 256 when applied to small datasets, which are characterized by a small number of classes). To streamline the size of the proposed model, the number of neurons was set to 128 specifically for smaller datasets.

4.7.3 Effect of the dropout value on each stream

The performance of the two-stream model with dropout is shown in Fig. 14. Both the temporal-LSTM and joint-motion modules show significant improvement as the dropout value is increased from zero to 0.3. The accuracy of the joint-motion module fluctuates within the range 61.6–69.58%, while that of the temporal LSTM module varies in the

Fig. 14 Accuracy for each stream on the JHMDB dataset for varying values of the dropout

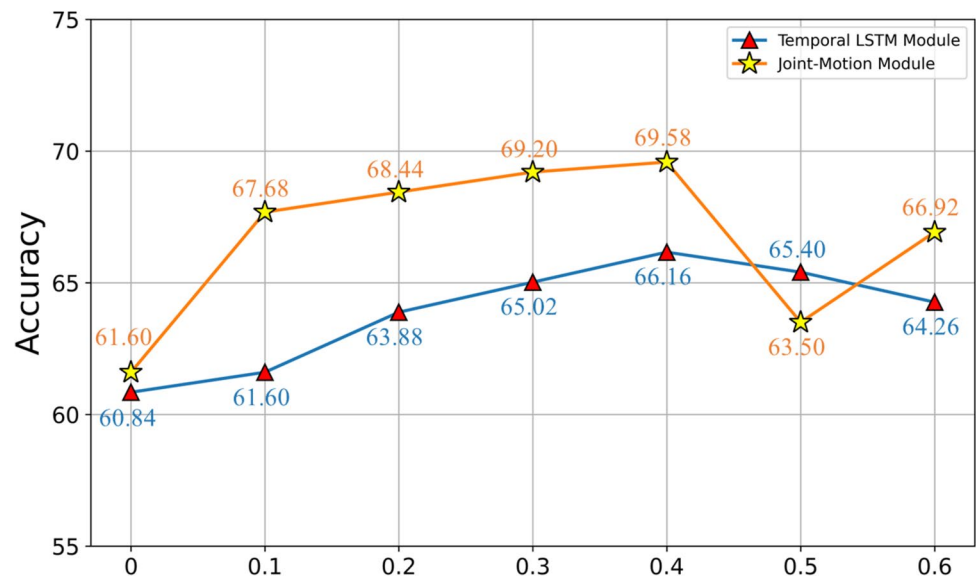


Table 10 Accuracy comparison between mean and max aggregation

Model	JHMDB	Florence-3D Action	SBU Interaction	Penn Action
Temporal-LSTM	61.15	95.35	89.09	91.18
Joint-motion	67.99	95.35	89.09	94.82
Proposed method with max aggregation	68.71	97.67	90.91	95.10
Proposed method with mean aggregation	68.35	97.67	85.45	95.10

range 60.84–66.16%. For values for the dropout of 0.4 to 0.6, the accuracies of both streams decrease minimally, with that of the joint-motion module dropping from 69.58 to 66.92%, and that of the temporal LSTM reduced by about 1.9%. Hence, the optimal dropout value for both streams is between 0.3 and 0.5. A dropout value of zero matches the original method. The proposed method (DSConVBlockV2) gave superior performance compared to the original method (DSConVBlock), with an improvement of approximately 7.97% when a dropout value of 0.4 was used.

4.7.4 Impact of mean and max aggregation

Table 10 presents a comparison of the effectiveness of two aggregation techniques, namely mean and max aggregation, on the accuracy of the proposed model. This evaluation was conducted using fold 2 of the four datasets. On the Florence-3D Action and Penn Action datasets, the accuracy of the proposed approach using max and mean aggregation was approximately 97.67% and 95.10%, respectively. For the JHMDB and SBU Interaction datasets, max aggregation gave higher accuracy than mean aggregation, with improvements of 0.36% and 5.46%, respectively. These outcomes lead us to the conclusion that the max aggregation technique yields superior performance for this scheme, and its adoption is advocated to attain optimal results.

5 Conclusion

In this work, a two-stream model has been introduced in which temporal LSTM and joint-motion modules are employed to address the vanishing gradient issue and improve the accuracy of spatial information. By combining the proposed method with skeletal data, the spatial-temporal features can be effectively extracted. Almost all of the experiments on four benchmark datasets demonstrated the superiority of the two-stream model with LSTM and DSConV over temporal LSTM alone. The joint-motion module outperformed temporal LSTM when a single skeleton input was provided (one person per video), whereas temporal LSTM excelled when multiple skeletons were input (two or more people per video). Two pre-trained models, MoveNet and Mediapipe, were also included to give real-time operation. One limitation arose when skeletons were highly similar, and the prediction accuracy was affected. The impact of the pre-trained models on the method's performance was stressed, and careful selection and enhancement are recommended to ensure the accuracy of skeleton estimation and to overcome the limitations of this approach. One limitation of the proposed method is the reduced prediction accuracy when dealing with highly similar skeletons. This issue highlights

the impact of pre-trained models on the overall performance of the method, underscoring the necessity for careful selection and enhancement of these models to ensure accurate skeleton estimation. Additionally, the performance of the proposed model is significantly influenced by the processing and normalization of input data. These factors contribute to potential biases and challenges in scaling the approach to other datasets, thereby affecting the generalizability of the model. In future work, methods combining RGB with optical flow will be explored, and local-global correlations will be analyzed to improve the background extraction and enhance the prediction accuracy of skeletal component motion. Additionally, the computational cost of the proposed model will be calculated and compared with state-of-the-art (SOTA) methods to demonstrate its suitability for real-time applications.

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Data availability The data and material presented in this study are available on request from the corresponding author.

Code availability The code that supports for this research are available on request from the corresponding author or the first author with email address such as cklu@ntnu.edu.tw or lehoangcongk16spkt@gmail.com, respectively.

Declarations

Competing interests All authors declare that they have no conflicts of interest.

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